

Ice Cream in a Bag!

Kitchen Science Fun





How Does Milk Become Ice Cream?

How does runny milk become scoopable ice cream?



Is milk solid or liquid?



Is ice cream solid or liquid?

Chotu the Ice Cube ❄️

Chotu gets super cold
with help from Captain
Salt!

Minnie the Milk Drop 🥛

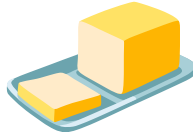
Minnie loves to flow and
swirl — until Chotu freezes
her into yummy ice cream!



What Are States of Matter?

Everything around us is either a solid or a liquid (or a gas)! Let's look at some examples from our very own Indian kitchen!



Solids 

Solids keep their shape! Like an Amul butter block or a steel Bowl — they don't flow or change shape on their own.



Liquids 

Liquids flow and take the shape of their container! Like milk poured from a Water Pot, or warm ghee melting on a hot paratha. Yummy!

Freezing and Melting Magic

❄️ Cooling turns liquids into solids!

When milk gets very cold, it becomes solid ice cream. That is freezing!

🔥 Warming turns solids into liquids!

When a solid gets warm, it melts back into a liquid. That is melting!

🍦 Example: Think about Kulfi in Delhi summer heat — it starts as a solid frozen treat, but melts quickly in the hot sun! And frozen Aam Kulfi lollies? They are solid mango pulp made icy cold!

Cooling = Liquid → Solid ❄️ | Warming = Solid → Liquid 💧



The Magic of Salt!



✓ Salt makes ice extra cold

When you add salt to ice, it melts a little — and that makes the ice even colder than before!

✓ Salt melts ice, then freezes it fast!

Salt lowers the freezing point of water. This super-cold ice can freeze our milk mix quickly and turn it into ice cream!

✓ Kulfi-walas know this trick!

In India, kulfi-walas pack metal moulds in ice mixed with rock salt to make kulfi freeze faster. Smart science!

Steps 1-3: Let's Begin!



Step 1 – Milk Mix

Pour $\frac{1}{2}$ cup milk, 1 tbsp sugar & 2 drops vanilla into small zip-lock pouch. Seal tight!

Step 2 – Ice Bag

Fill big pouch halfway with ice cubes. Add 4-5 tbsp salt.
Ready!

Step 3 – Combine!

Put small milk pouch inside big ice-salt pouch. Seal big pouch tightly!



Steps 4, 5 & 6

Open!

Open outer bag carefully.
Wipe salt water off small bag.



Shake!

Wear gloves or wrap bag in a towel. Shake for 5 minutes!



Scoop!

Scoop out your solid ice cream! Look
– milk became ice cream!



Taste Time!

Everyone takes their first bite together! Yummy! We made ice cream with science!

What Did Captain Salt Do?



✓ Captain Salt made ice extra cold!

Salt mixes with ice and lowers its temperature, making it super cold — cold enough to freeze our milk mix!

✓ Captain Salt helped freeze milk fast!

The salty ice pulled heat out of the milk mix very quickly, turning the liquid milk into solid, scoopable ice cream!

✓ Chotu and Minnie say thank you!

Without Captain Salt, Chotu the Ice Cube couldn't freeze Minnie the Milk Drop. Together they made yummy ice cream!



Try It at Home! 🍦 Make ice cream with
Rooh Afza or Badam Milk!

⚠️ **Do it with a parent!**
Thank you & see you next
time! 🎉